Reviewer Pack — Minimal Symmetry Group Order and Monotone Circuit Lower Boun…

Entry 88307033055a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Symmetry Group Order and Monotone Circuit Lower Bounds for k-CLIQUE

Entry ID: `88307033055a` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-22 02:41:59 UTC

1. Conjecture

Field A (mathematical branch): Group Theory: Symmetry Groups **Field B** (complexity object): Complexity Theory: Monotone Circuit Complexity for k-CLIQUE

Statement:

> ['For any monotone circuit family C_k computing the k-CLIQUE problem, there exists a nontrivial symmetry group G of size Θ(n^k) such that all circuits in C_k admit an isomorphic copy with G as its automorphism group.', 'Furthermore, for every instance of n variables, the minimal order of a symmetry group of any circuit in C_k is at least 2^n.', 'If the k-CLIQUE problem has monotone circuits of size n^{f(n)}, then there exists an explicit nontrivial symmetry group G with order at least $2^{\{n}$k} such that all circuits in C_k admit a copy with G as its automorphism group.']

Rationale (proposer's reasoning):

> ['Symmetry groups can capture the inherent structure and redundancy in computational problems. If the minimal symmetry group order of k-CLIQUE circuits grows exponentially with n, it would suggest an intrinsic complexity to the problem that is not easily mitigated by symmetry-breaking techniques.', 'This conjecture connects the rich theory of symmetry groups with the study of monotone circuits, potentially offering new insights into the difficulty of the k-CLIQUE problem and leading to nontrivial lower bounds.', 'The exponential growth in group order would imply a fundamental barrier to circuit minimization for k-CLIQUE that goes beyond current understanding.']

Taxonomy category: `MONOTONE_CLIQUE` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `55d85c2034774451`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the k-CLIQUE problem, if a monotone circuit C_k has size n^{f(n)} and its minimal symmetry group order is at least 2^n for all circuits in C_k, then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	UNCERTAIN	0.90	HITS	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "symmetry group" AND "monotone circuit complexity" AND k-CLIQUE" - "automorphism group" AND "k-CLIQUE problem" AND minimal order" - "nontrivial symmetry group" AND monotone circuits AND θ(n^k) size

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.5s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_monotone_circuit(n, k):
        # Placeholder for monotone circuit generation logic
        return [random.choice([0, 1]) for _ in range(n**k)]

    def calculate_symmetry_group(circuit):
        # Placeholder for symmetry group calculation logic
        return set()

    n = random.randint(5, 40)
    k = random.randint(2, min(3, n))
    circuit = generate_monotone_circuit(n, k)
    G = calculate_symmetry_group(circuit)

    if not G:
        return {
            "metric_name": "symmetry_group_order",
            "metric_value": 0,
            "instances_tested": 1,
            "conjecture_holds": False,
```

```
            "counterexample": "mapping_undefined"
        }

    symmetry_group_order = len(G)
    if symmetry_group_order < 2**n:
        return {
            "metric_name": "symmetry_group_order",
            "metric_value": symmetry_group_order,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": f"Symmetry group order {symmetry_group_order} is less than 2^n for n
        }

    return {
        "metric_name": "symmetry_group_order",
        "metric_value": symmetry_group_order,
        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2**i - 1 for i in range(5, 35)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample = next((r["counterexample"] for r in results if r["conjecture_holds"]), "")
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_fa
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
 'mapping_undefined'}
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
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TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
```

```
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'symmetry_group_order', 'metric_value': 0, 'instances_tested': 1, 'conjecture

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae5ade61.py", line 79, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae5ade61.py", line 79, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                ~^^^^^^^^
KeyError: 'seed'
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which prevents us from evaluating the conjecture's validity. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14590
2	propose	ollama_remote	glm4:latest	0	0	15632
3	preregistration	ollama_remote	glm4:latest	0	0	11049
4	novelty	ollama_remote	glm4:latest	0	0	8427
5	novelty	ollama_remote	glm4:latest	0	0	10407
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12800
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6585
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12342
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9155
10	judge	ollama_remote	glm4:latest	0	0	27363

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 128349 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/88307033055a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/88307033055a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/88307033055a.tar.gz` (if generated)